

which player's state according to the game-world is updated, taking into consideration of the world-restrictions.

- ▶ World related data updates which include passive elements which are static and logic-based elements which are dynamic.
- ▶ AI-based updates which include updating goals and states of NPCs, update the decision engine and the game world according to the change in player's state.

Approaches of updating game-Loop

There are three approaches that are followed to update the game loop:

i) To design update/render the process in a single loop

Advantages

- ▶ Both routines are given equal importance.
- ▶ Logic and presentation are properly coupled.

Disadvantages

- ▶ Variation in complexity of one of the routines will influence the other.
- ▶ There is no control over how often a routine is updated.

ii) To Design update process using two threads with decoupled frequency

Advantages

- ▶ Both update and render loops at the same time.

Disadvantages

- ▶ Not all machines are that good at handling threads. As a single CPU has to handle it, precise timing problems can occur. Synchronization issues, as two threads accessing same data

iii) To design an update/render single threaded decoupled process

Advantages

- ▶ It has better control than thread and has simpler programming to do.
- ▶ No need of sharing and synchronizing.

Disadvantages

- ▶ It assumes that the timer takes 0 sec. to complete.
- ▶ It does not handle Alt-Tab scenario.
- ▶ There is no nesting with
- ▶ different frequencies.